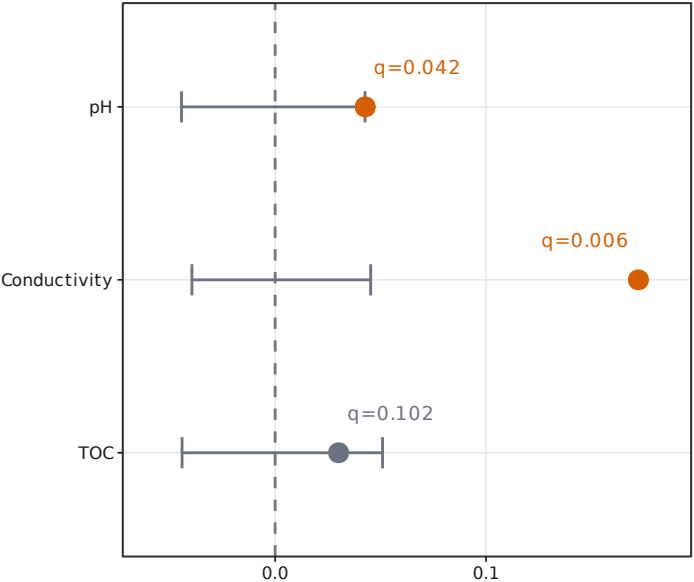


Assembly-process labels need assumption and preprocessing gates

A. Short-range niche signal gate

Observed rho with 95% label-permutation interval

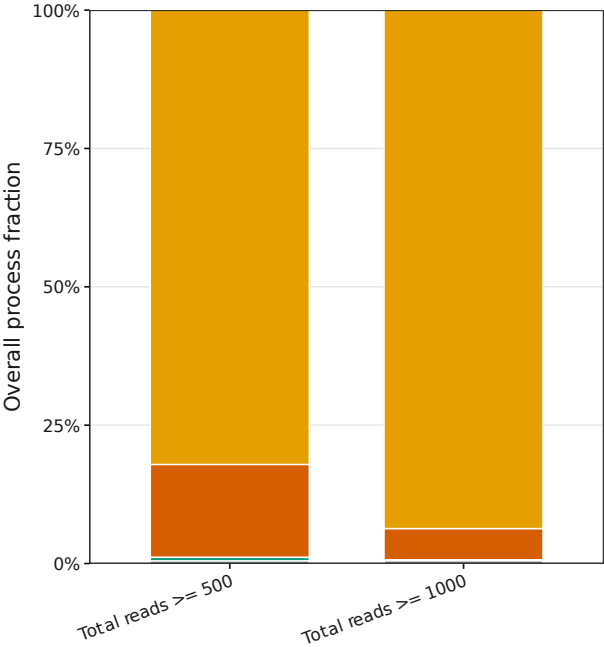


Phylogenetic distance vs niche difference rho

BH supported ● FALSE ● TRUE

B. Feature-filter sensitivity

Pairwise label agreement = 81.4%



The primary analysis covers 47.4% of reads; rare-biosphere processes are outside its estimand.